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ALEXEY VELASQUEZ

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Data Scientist – ML Engineer

TECHNICAL SKILLS

Languages Python, SQL.

Platforms PostgreSQL, VS Code, N8N, Azure, Databricks, MLflow, Docker, Git, Power BI.

ML / AI scikit-learn, PyCaret, LightGBM, XGBoost, PyTorch/Torch, TensorFlow, Generative AI / LLMs (Ollama).

Data NumPy, pandas, SciPy, Matplotlib, Seaborn.

Methods SHAP, RAG, statistical testing, A/B testing.

EXPERIENCE

Data Scientist

Jan 2026 - mar 2026

Cactus Home Offer · cactushomeoffer.com

Florida, USA, Remote

- Built an end-to-end speech intelligence pipeline combining Transformer-based speech recognition, scikit-learn-based pattern detection, and SHAP-based interpretability to generate per-segment behavioral insights from real estate sales calls, achieving up to 80% reduction in manual call review time and measurably improving closer performance.
- Replaced rule-based KPI extraction with a context-aware Gen AI (LLM) to detect objections, buying signals, rapport quality, and call outcome from full transcripts in mixed Spanish/English, eliminating keyword logic.
- Integrated the pipeline with n8n for CRM-triggered real-time inference, leveraging Azure-hosted services and Databricks API integrations to automate sales coaching.

Data Scientist

Jun 2025 - Dec 2025

Luis Buys Homes LLC

Florida, USA, Remote

- Built an end-to-end NLP and Sentiment Analysis (Opinion Mining) pipeline containerized with Docker using Transformer-based Neural Networks (BERT) to analyze call transcriptions from first-contact home sales interactions, identifying behavioral and conversational patterns correlated with successful deals.
- Designed, developed, and deployed an automated standalone data processing interface (GUI/Demo) for sentiment analysis, opinion mining, and trend tracking in n8n.
- Delivered real-time analytics and decision-support dashboards by connecting ML outputs stored in PostgreSQL to Power BI, supporting data-driven sales strategy optimization.

Machine Learning Engineer

Aug 2023 – Aug 2025

EIA University

Medellin - Col

- Developed a pattern recognition and anomaly detection pipeline using LightGBM and XGBoost to identify faults in structural health monitoring systems for air turbines, achieving 99.5% detection accuracy.
- Conducted comparative model evaluation (A/B testing) across multiple machine learning approaches, including Random Forest, LightGBM, Support Vector Machines, and Autoencoders, selecting optimal models based on performance metrics and use-case requirements.
- Designed and managed PostgreSQL databases to store large-scale monitoring data, enabling continuous real-time data ingestion and ML pipeline execution throughout the project lifecycle.

EDUCATION

Master's in Engineering (Machine Learning Focus): 1st Class (94%), EIA University.

Aug 2025